

CSE 318 Assignment-04

Decision Tree

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Subsection: c2

1 Introduction

This study implements a decision tree classifier to evaluate and compare three splitting criteria—Information Gain (IG), Information Gain Ratio (IGR), and Normalized Weighted Information Gain (NWIG)—on the Iris and Adult datasets. IG quantifies the reduction in entropy from a split, IGR adjusts IG by the attribute’s intrinsic information to reduce bias, and NWIG introduces a penalty based on attribute cardinality and dataset size, which may yield negative gains and cause early stopping.

The evaluation consisted of the following:

- Performing 20 random 80/20 train-test splits on each dataset to compute average accuracy at a maximum depth of 20.
- Comparing average accuracy across tree depths from 0 to 4.
- Analyzing tree complexity in terms of node count and depth, without applying pruning.

2 Results

2.1 Accuracy Across Maximum Depths

Table 1: Average Accuracy Across Maximum Depths for Iris Dataset

Max Depth	IG	IGR	NWIG
0	94.17	95.17	95.50
1	95.50	94.00	94.17
2	94.83	94.50	96.33
3	92.50	95.33	94.67
4	94.00	94.17	94.00

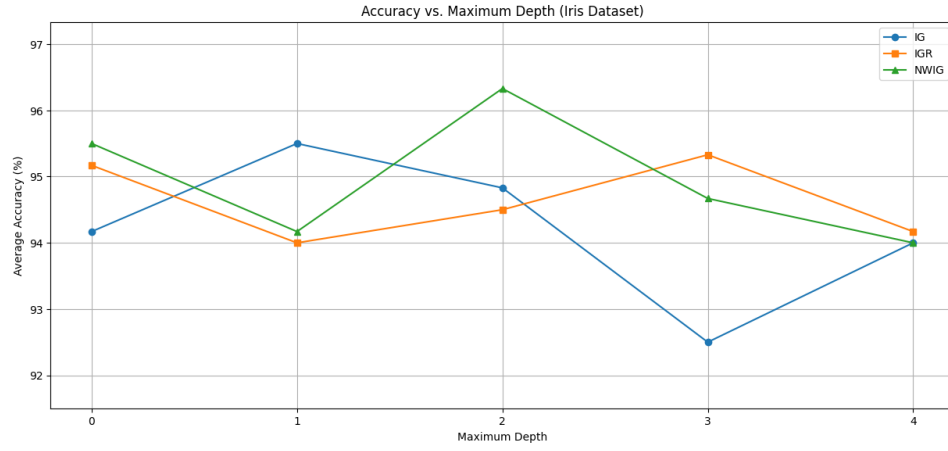


Figure 1: Accuracy trends for Iris dataset across maximum depths.

Table 2: Average Accuracy Across Maximum Depths for Adult Dataset

Max Depth	IG	IGR	NWIG
0	81.91	82.15	82.91
1	75.87	79.97	76.02
2	82.62	80.12	82.69
3	83.75	84.00	84.17
4	83.82	84.27	84.08

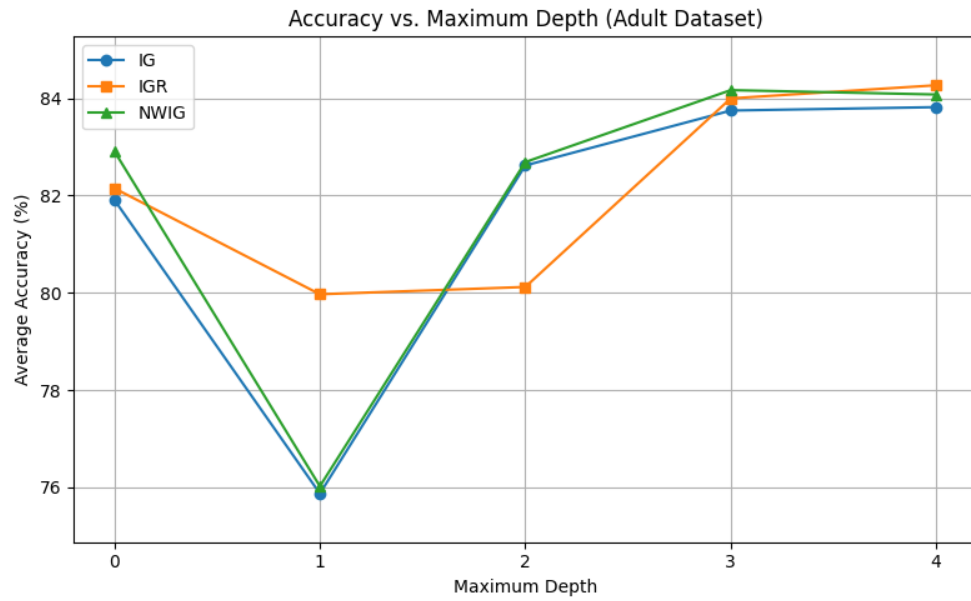


Figure 2: Accuracy trends for Iris dataset across maximum depths.

3 Discussion

- **Accuracy Trends:** IGR consistently outperforms IG and NWIG, particularly in the more complex Adult dataset, while the simpler Iris dataset yields high accuracy across all criteria with less variation.
- **Tree Complexity:** NWIG results in significantly smaller trees due to its penalty on high-cardinality attributes, helping reduce overfitting—especially in large datasets like Adult.
- **Depth Impact:** Accuracy tends to improve with tree depth up to a point (typically depth 3–5), after which overfitting may occur, as seen with NWIG in the Iris dataset.

4 Conclusion

IGR consistently achieved the highest accuracy across both datasets, making it the most reliable criterion overall. NWIG produced smaller trees, which is beneficial for complex datasets like Adult, though it showed occasional instability. IG was the simplest but often led to larger trees and potential overfitting. Overall, IGR offers the best trade-off between accuracy and tree complexity.